

Automated Cross-Sell Engine

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What it does

Vrtly screens already sit in aesthetics waiting rooms, treatment rooms, and on patient phones via Info Packs. The EMR integration turns them into a revenue layer. They know who's in the clinic, where they are, and what each patient has bought — modeled against every other practice on Vrtly to predict what they'll buy next.

Vrtly overlays media on those paths — right ad, right patient — and models lifetime value to grow practice revenue short-term and long.

The flow

- 0. Clean up the service catalog.** Practices name the same treatment three different ways — *Botox 20u forehead*, *BTX glabella*, *wrinkle relaxer upper face*. We map every practice's menu onto one proprietary shared list. The more practices join, the smarter the mapping gets. (Detailed below.)
- 1. Find the buying patterns.** We read every booking and invoice and surface the practice's real cross-sell paths — *Injectables* → *Skin Care* · 96 guests · avg \$1,063 second visit · 40d between. This is a novel insight for most practices based on our product UX panels.
- 2. Practice picks goals.** What to grow — *fill Tuesday Botox*, *get Injectables guests into Skin Care*, *win back 90-day-lapsed VIPs*.
- 3. Smart playlists.** A propensity-to-buy model¹ picks content for each day's patients based on the practice's goals and patterns. Playlists build themselves.
- 4. Clinic-aware screens.** Screens know the day's schedule — who's in the lobby, who's in which room, what they came in for — and shift content to match. The Scheduler view is currently live in our prototype.
- 5. Attribution. Proof it worked.** Every patient who saw a piece of content and then bought it gets matched back. The screens get credit for the sales they drove.

Zenoti is the first EMR connector. Live at two centers, ~5,000 appointments cached, ~\$1.3M under management. Working prototype at prototypes.vrtly.ai. Provisional patent filed.

API	Endpoint	What we use it for
Appointments	GET /v1/appointments	Core dataset. Feeds /insights, /analytics/paths, /ltv, /retention, /clinic/schedule, /clinic/media-plan, /attribution/*, and /analyst/chat.
Invoice Detail	GET /v1/invoices/{invoice_id}	Service-level revenue. Powers /ltv (Pareto, segment LTV), revenue-weighted cross-sell paths, and at-risk patterns.
Service Catalog	GET /v1/centers/{center_id}/services	Full service menu per center. Feeds the classifier and every category-aware endpoint.
Business Summary	POST /v1/reports/business_summaries	QA harness. Reconciles aggregates against Zenoti's report.
Centers	GET /v1/centers	One-time discovery. Not in the runtime hot path.

How the classifier works

Step 0 in detail. Five stages, designed to spend as little model budget as possible.

- 1. Ingest.** Pull raw service strings from Zenoti.
- 2. Match.** Exact and fuzzy match against Vrtly's proprietary canonical taxonomy plus this practice's prior confirmations and network-wide aliases. Most hits resolve here, no extra modeling needed.
- 3. Custom classification with Haiku 4.5.** Residuals hit a customized Claude Haiku 4.5 with the practice's confirmed mappings as in-context examples. Returns category, confidence, alternates.
- 4. Human review.** Sub-threshold cases land in a one-click y/n UI. Every confirmation persists per-practice and propagates network-wide.
- 5. Nightly enrichment.** An agent searches Google nightly on stubborn unknowns and returns more context to the modeling layer. Experiments show ~40% increase in classification across the long tail.

¹ Gradient-boosted decision trees (XGBoost / LightGBM).